ukka-venugopalam.github.io

Page Replacement Simulator & Analysis

Concepts & Time Complexity

FIFO (First-In, First-Out)

Concept: Replaces the oldest page in memory.

GATE Note: Suffers from *Belady's Anomaly*. Check the line graph below to see it!

Time Complexity: O(1) per step.

LRU (Least Recently Used)

Concept: Replaces the page not used for the longest time.

GATE Note: Stack algorithm: guaranteed to never suffer from Belady's Anomaly.

Time Complexity: O(N) per step.

OPT (Optimal)

Concept: Replaces the page that will not be used for the longest time in the future.

GATE Note: Yields the theoretical minimum faults. Impossible to implement practically.

Time Complexity: O(N) per step.

Simulation Settings

Try Example (GATE)

Algorithm Selection Mode:

Compare All (FIFO, LRU, OPT)

Reference String (comma or space separated):

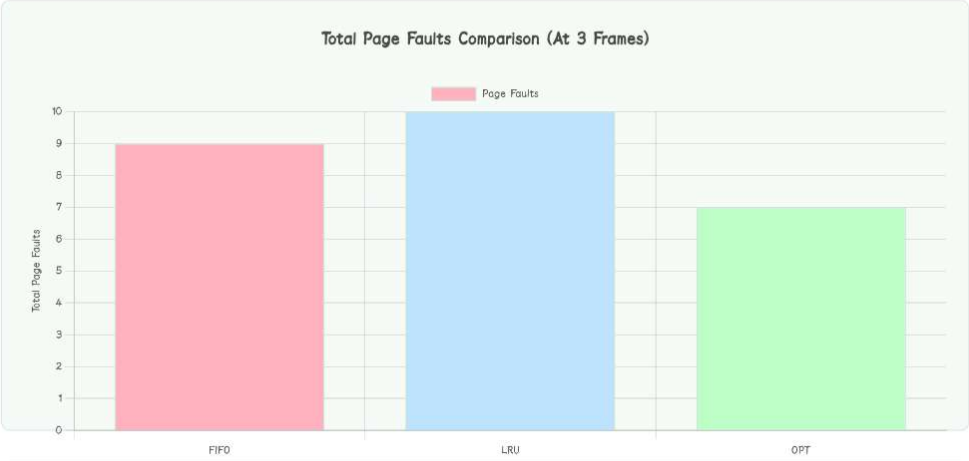
1 2 3 4 1 2 5 1 2 3 4 5

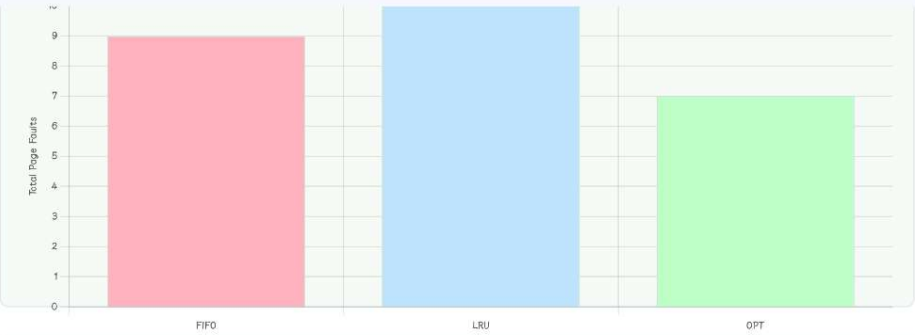
Target Frames (For Tables): 3

Max Frames for Graph (Belady's Demo): 7

Run Simulation & Generate Graphs

Data Visualization





Summary (At 3 Frames)

Algorithm	Page Faults	Page Hits	Hit Rate	Fault Rate
FIFO	9	3	25.00%	75.00%
LRU	10	2	16.67%	83.33%
OPT	7	5	41.67%	58.33%

🏆 **Best Algorithm: OPT (Page Faults = 7)**

Page Fault (Miss) Page Hit

FIFO Simulation

Ref	1	2	3	4	1	2	5	1	2	3	4	5
F1	1	1	1	4	4	4	5	5	5	5	5	5
F2	–	2	2	2	1	1	1	1	1	3	3	3
F3	–	–	3	3	3	2	2	2	2	2	4	4
Status	✗	✗	✗	✗	✗	✗	✗	✓	✓	✗	✗	✓

▼ View Step-by-Step Explanation

Step 1: Request Page **1**. **FAULT!** Added to memory. Replaced via FIFO pointer at Frame 1.

Step 2: Request Page **2**. **FAULT!** Added to memory. Replaced via FIFO pointer at Frame 2.

Step 3: Request Page **3**. **FAULT!** Added to memory. Replaced via FIFO pointer at Frame 3.

Step 4: Request Page **4**. **FAULT!** Evicted Page 1. Replaced via FIFO pointer at Frame 1.

Step 5: Request Page **1**. **FAULT!** Evicted Page 2. Replaced via FIFO pointer at Frame 2.

Step 6: Request Page **2**. **FAULT!** Evicted Page 3. Replaced via FIFO pointer at Frame 3.

Step 7: Request Page **5**. **FAULT!** Evicted Page 4. Replaced via FIFO pointer at Frame 1.

LRU Simulation

Ref	1	2	3	4	1	2	5	1	2	3	4	5
F1	1	1	1	4	4	4	5	5	5	3	3	3
F2	–	2	2	2	1	1	1	1	1	1	4	4
F3	–	–	3	3	3	2	2	2	2	2	2	5
Status	✗	✗	✗	✗	✗	✗	✗	✓	✓	✗	✗	✗

▶ View Step-by-Step Explanation

OPT Simulation

Ref	1	2	3	4	1	2	5	1	2	3	4	5
F1	1	1	1	1	1	1	1	1	1	3	4	4
F2	–	2	2	2	2	2	2	2	2	2	2	2
F3	–	–	3	4	4	4	5	5	5	5	5	5
Status	✗	✗	✗	✗	✓	✓	✗	✓	✓	✗	✗	✓

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Time Complexity: $O(1)$ per step.

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Concept: Replaces the page not used for the longest time.

GATE Note: Stack algorithm: guaranteed to never suffer from Belady's Anomaly.

Time Complexity: $O(N)$ per step.

OPT (Optimal)

Concept: Replaces the page that will not be used for the longest time in the future.

GATE Note: Yields the theoretical minimum faults. Impossible to implement practically.

Time Complexity: $O(N)$ per step.

Simulation Settings

Try Example (GATE)

Algorithm Selection Mode:

Run Single Algorithm

Algorithm 1:

FIFO

Reference String (comma or space separated):

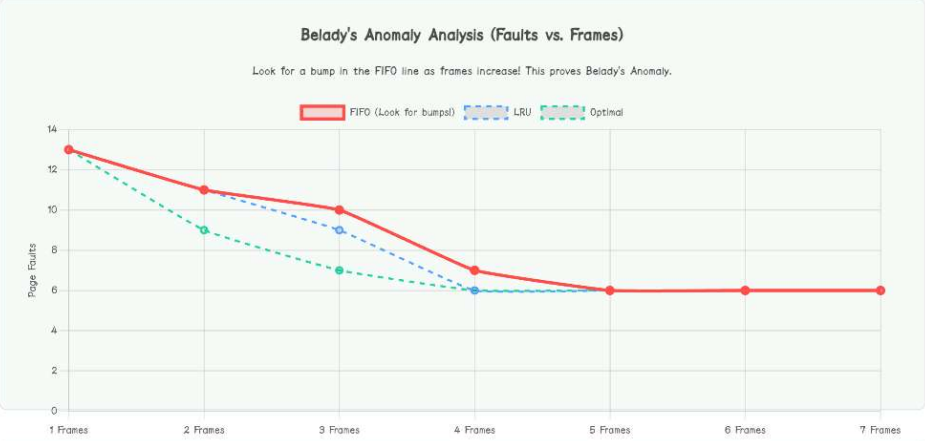
7 0 1 2 0 3 0 4 2 3 0 3 2

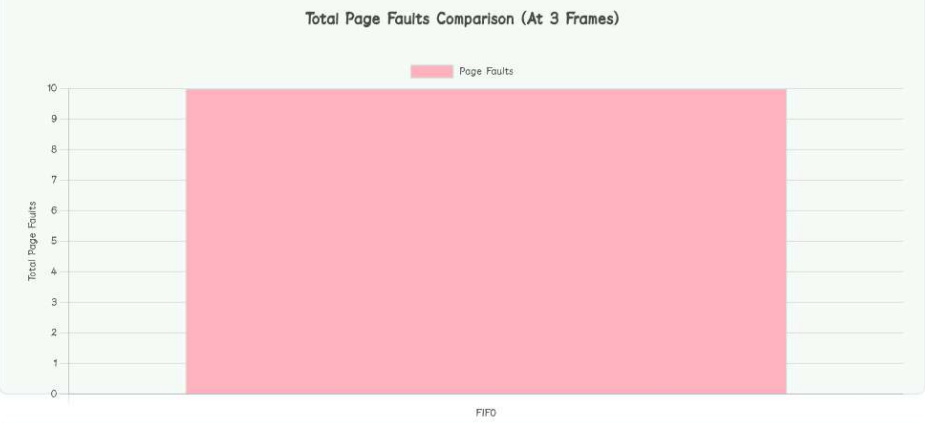
Target Frames (For Tables): 3

Max Frames for Graph (Belady's Demo): 7

Run Simulation & Generate Graphs

Data Visualization





Summary (At 3 Frames)

Algorithm	Page Faults	Page Hits	Hit Rate	Fault Rate
FIFO	10	3	23.08%	76.92%

Try Example (GATE)

Algorithm Selection Mode:

Run Single Algorithm

Algorithm 1:

FIFO

Reference String (comma or space separated):

7 0 1 2 0 3 0 4 2 3 0 3 2

Target Frames (For Tables):

3

Max Frames for Graph (Belady's Demo):

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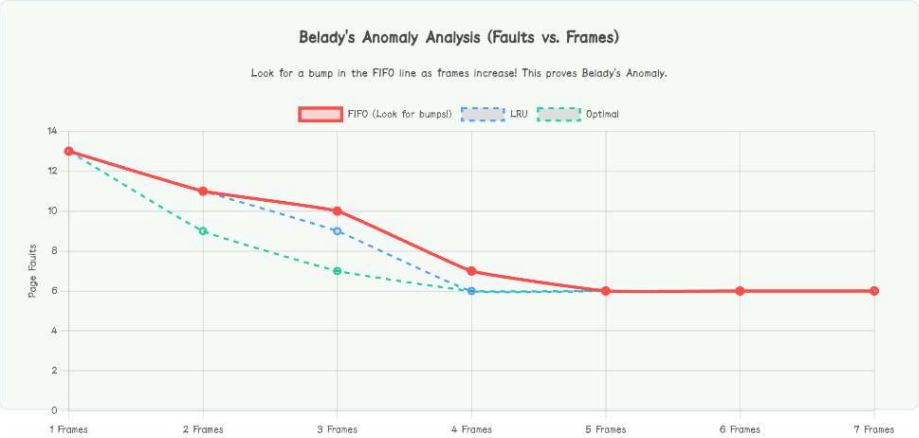
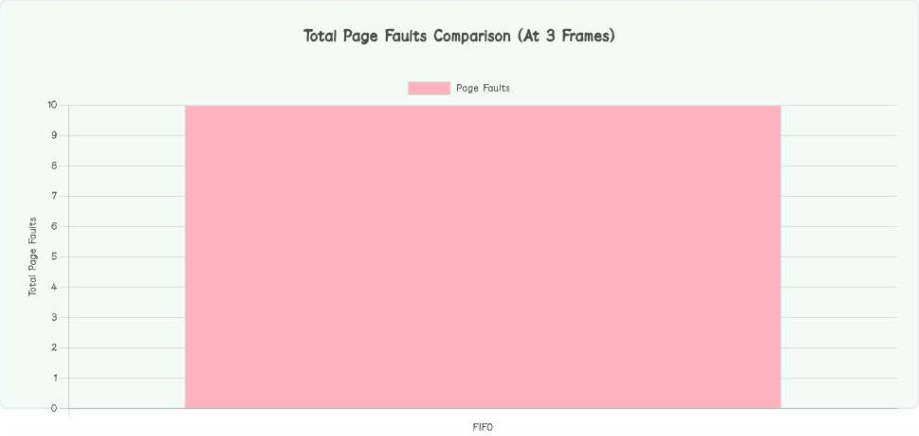
Run Simulation & Generate Graphs

Data Visualization





Summary (At 3 Frames)

Algorithm	Page Faults	Page Hits	Hit Rate	Fault Rate
FIFO	10	3	23.08%	76.92%

Page Fault (Miss) Page Hit

FIFO Simulation

Ref	7	0	1	2	0	3	0	4	2	3	0	3	2
F1	7	7	7	2	2	2	2	4	4	4	0	0	0
F2	-	0	0	0	0	3	3	3	2	2	2	2	2
F3	-	-	1	1	1	1	0	0	0	3	3	3	3
Status	✗	✗	✗	✗	✓	✗	✗	✗	✗	✗	✗	✓	✓

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Concept: Replaces the page not used for the longest time.

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Time Complexity: O(N) per step.

OPT (Optimal)

Concept: Replaces the page that will not be used for the longest time in the future.

GATE Note: Yields the theoretical minimum faults. Impossible to implement practically.

Time Complexity: O(N) per step.

Simulation Settings

Try Example (GATE)

Algorithm Selection Mode:

Compare 2 Algorithms

Algorithm 1: FIFO

Algorithm 2: LRU

Reference String (comma or space separated):

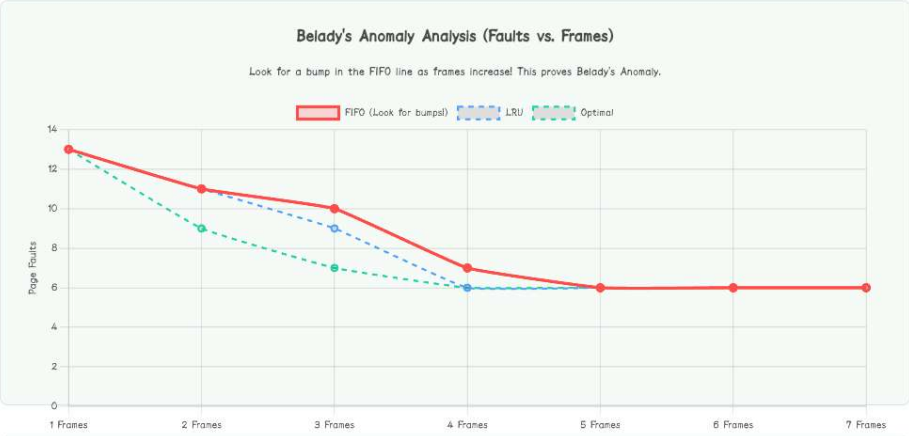
7 0 1 2 0 3 0 4 2 3 0 3 2

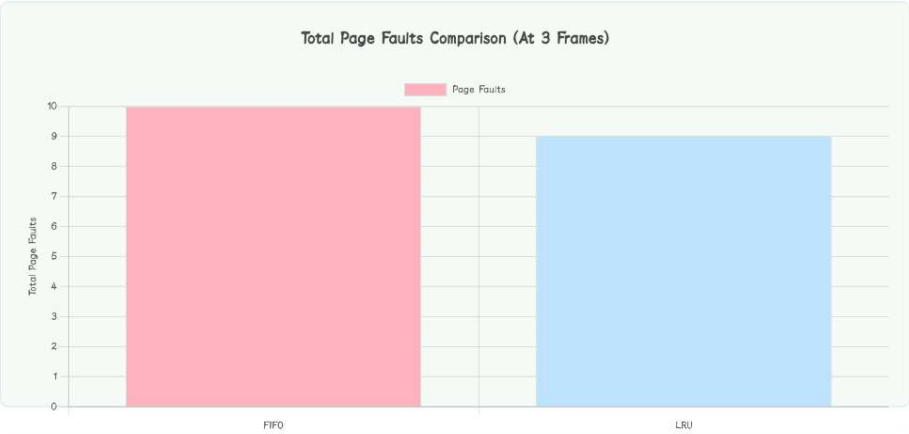
Target Frames (For Tables): 3

Max Frames for Graph (Belady's Demo): 7

Run Simulation & Generate Graphs

Data Visualization

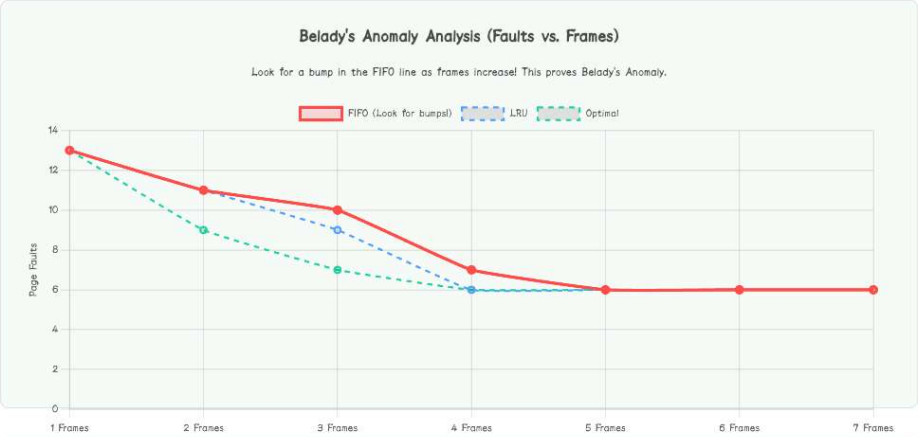


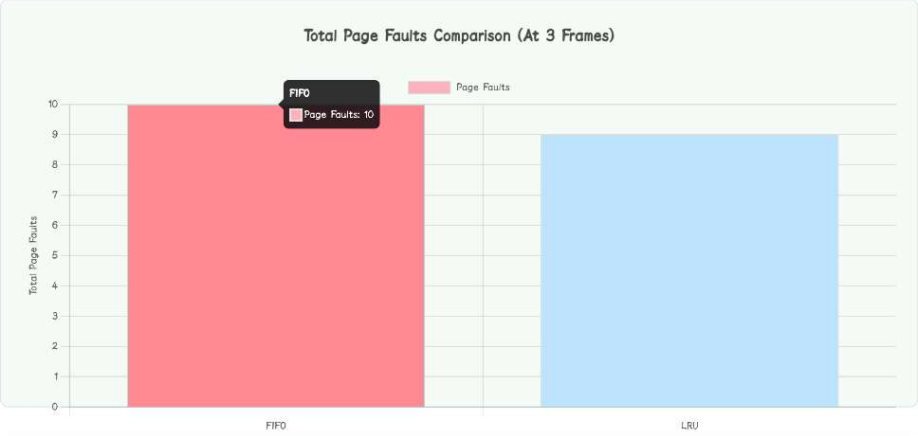


Summary (At 3 Frames)

Algorithm	Page Faults	Page Hits	Hit Rate	Fault Rate
FIFO	10	3	23.08%	76.92%
LRU	9	4	30.77%	69.23%

Data Visualization





Summary (At 3 Frames)

Algorithm	Page Faults	Page Hits	Hit Rate	Fault Rate
FIFO	10	3	23.08%	76.92%
LRU	9	4	30.77%	69.23%

🏆 **Best Algorithm: LRU (Page Faults = 9)**

Page Fault (Miss) Page Hit

FIFO Simulation

Ref	7	0	1	2	0	3	0	4	2	3	0	3	2
F1	7	7	7	2	2	2	2	4	4	4	0	0	0
F2	–	0	0	0	0	3	3	3	2	2	2	2	2
F3	–	–	1	1	1	1	0	0	0	3	3	3	3
Status	✗	✗	✗	✗	✓	✗	✗	✗	✗	✗	✗	✓	✓

▶ View Step-by-Step Explanation

LRU Simulation

Ref	7	0	1	2	0	3	0	4	2	3	0	3	2
F1	7	7	7	2	2	2	2	4	4	4	0	0	0
F2	–	0	0	0	0	0	0	0	0	3	3	3	3
F3	–	–	1	1	1	3	3	3	2	2	2	2	2
Status	✗	✗	✗	✗	✓	✗	✓	✗	✗	✗	✗	✓	✓

▶ View Step-by-Step Explanation